



Refining the Sunspot Number Series to ISN V3

Challenges and Benefits for the Space Climate Community

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Abstract

The International Sunspot Number (ISN) remains affected by historical scale jumps such as the *Waldmeier break*, even after two major recalibrations. FARSuN (Findability and Accessibility of historical Raw Sunspot Numbers) tackles this by rebuilding the record from the ground up: collecting handwritten logs and tables from 1607–1980, digitizing them, and applying consistent metadata and validation. Early 2025 milestones include integrating the full 1610–1918 *Mittheilungen* corpus, digitizing more than 95% of the 2000+ Zürich observation tables, extracting 338 drawings (1056 entries) from the C. H. Adams notebooks, and launching quality-control pipelines for the Gruithuisen manuscripts. The reconstructed ISN V3 will deliver traceable uncertainties, hemispheric indices, and open FAIR-compliant access, strengthening space climate studies, solar cycle prediction, and coupled Sun–Earth research.

Project Pulse



Why Reconstruction?

The ISN inherits observer-dependent scale changes (for example the 1947 "Waldmeier jump") that cannot be removed by tweaking coefficients alone. FARSuN pioneers a full reconstruction that replays the entire observing chain.

- **Recalibration** only rescales an existing index; legacy gaps, missing metadata, and hidden assumptions stay baked in.
- **Reconstruction** retrieves the primary artefacts (drawings, tabular logs, weather notes) and recomputes the index with transparent uncertainty tracking.
- Direct access to the raw sources lets us audit outliers, model observer response curves, and issue reproducible releases such as ISN V3.

Recalibration vs Reconstruction

Legacy recalibration	FARSuN reconstruction
Linear scaling of composite series; no new metadata; residual biases (e.g. "Waldmeier jump") remain hidden.	Primary sources re-digitised with observer context, transparent uncertainty budgets, and reproducible release tooling.

Based on FARSuN interactive report section "The Problem" & "The Solution".

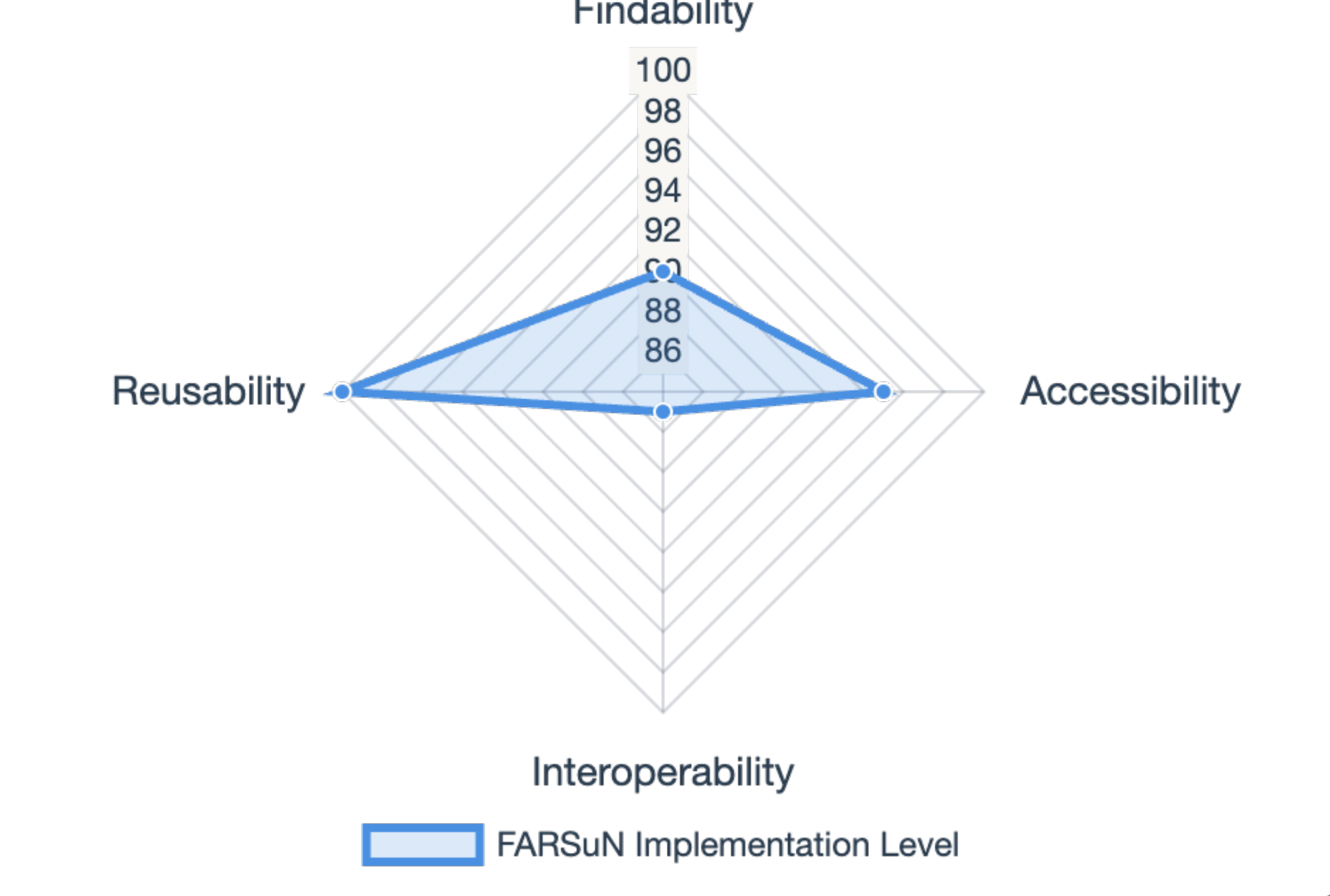
Four-Axis Methodology

- Gather Sources:** track down dispersed observing logs (Wolf's *Mittheilungen*, Zürich tables, Gruithuisen manuscripts, Adams drawings) and capture provenance metadata at ingest.
- Process Data:** combine handwriting/OCR engines (Transkribus models, custom scripts) with manual double-keying for fragile entries.
- Validate & Standardize:** reconcile aliases, observer IDs, and calendars; model group/spot counting behaviour to derive calibration curves.
- Disseminate:** publish machine-readable products, APIs, and VO-ready services with versioned uncertainty bundles for each release.

FAIR Principles

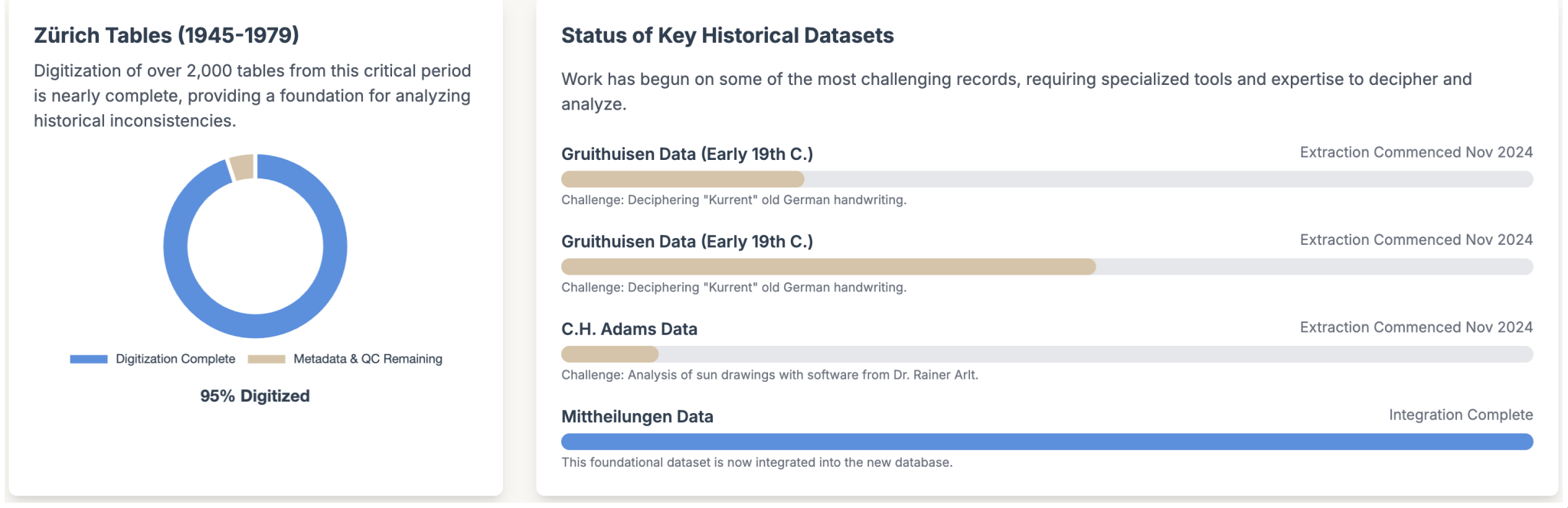
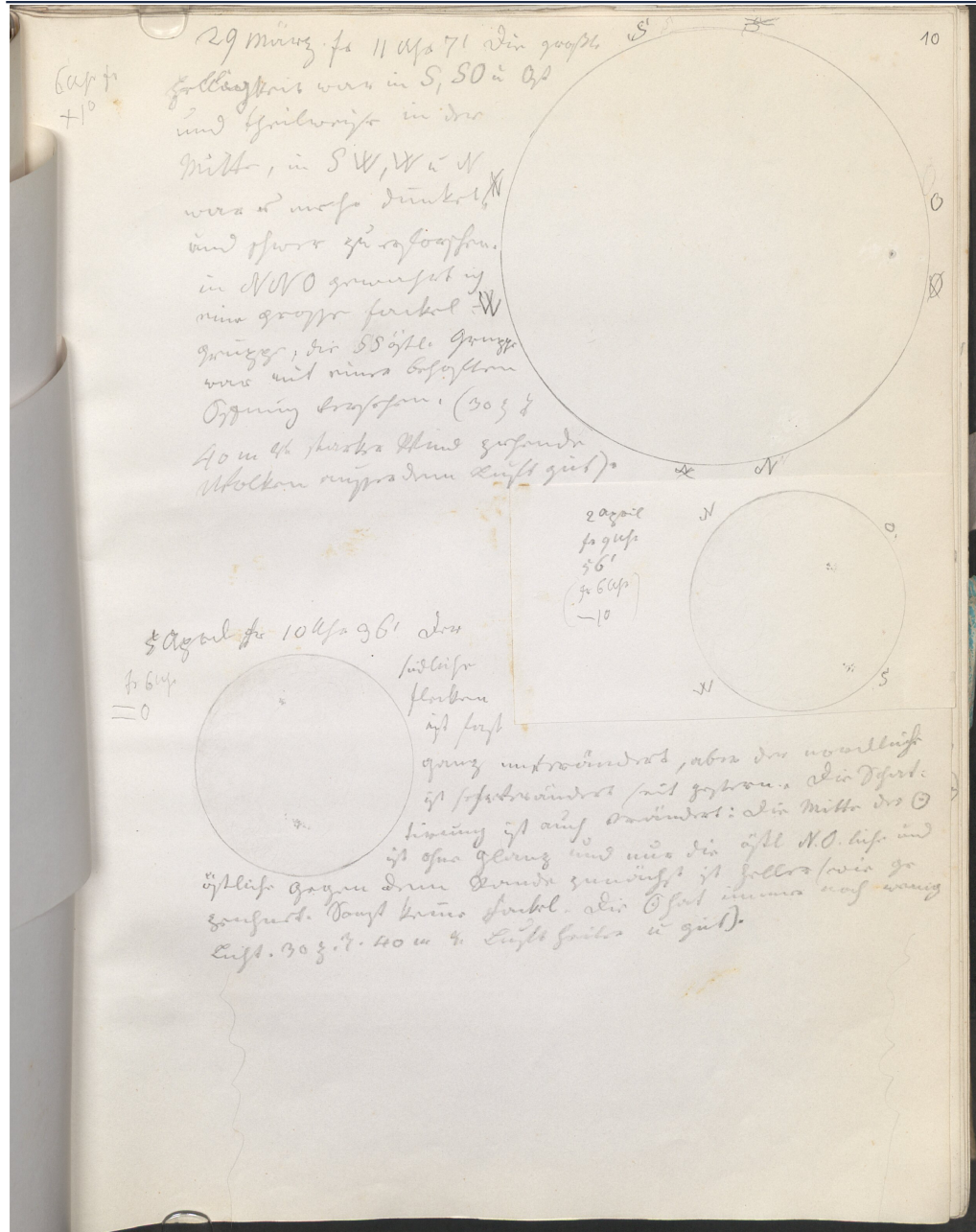
Every dataset and release roadmap is benchmarked against FAIR targets that we monitor with the dashboard shown below.

- **Findable** – persistent DOIs, rich observation-level metadata, and navigable catalogues.
- **Accessible** – open formats (CSV, NetCDF, VO tables) served via HTTPS and scripted download recipes.
- **Interoperable** – shared vocabularies, observer IDs, and VEO-ready semantics.
- **Reusable** – versioned releases with uncertainty models, provenance trails, and detailed caveats.



Key Datasets

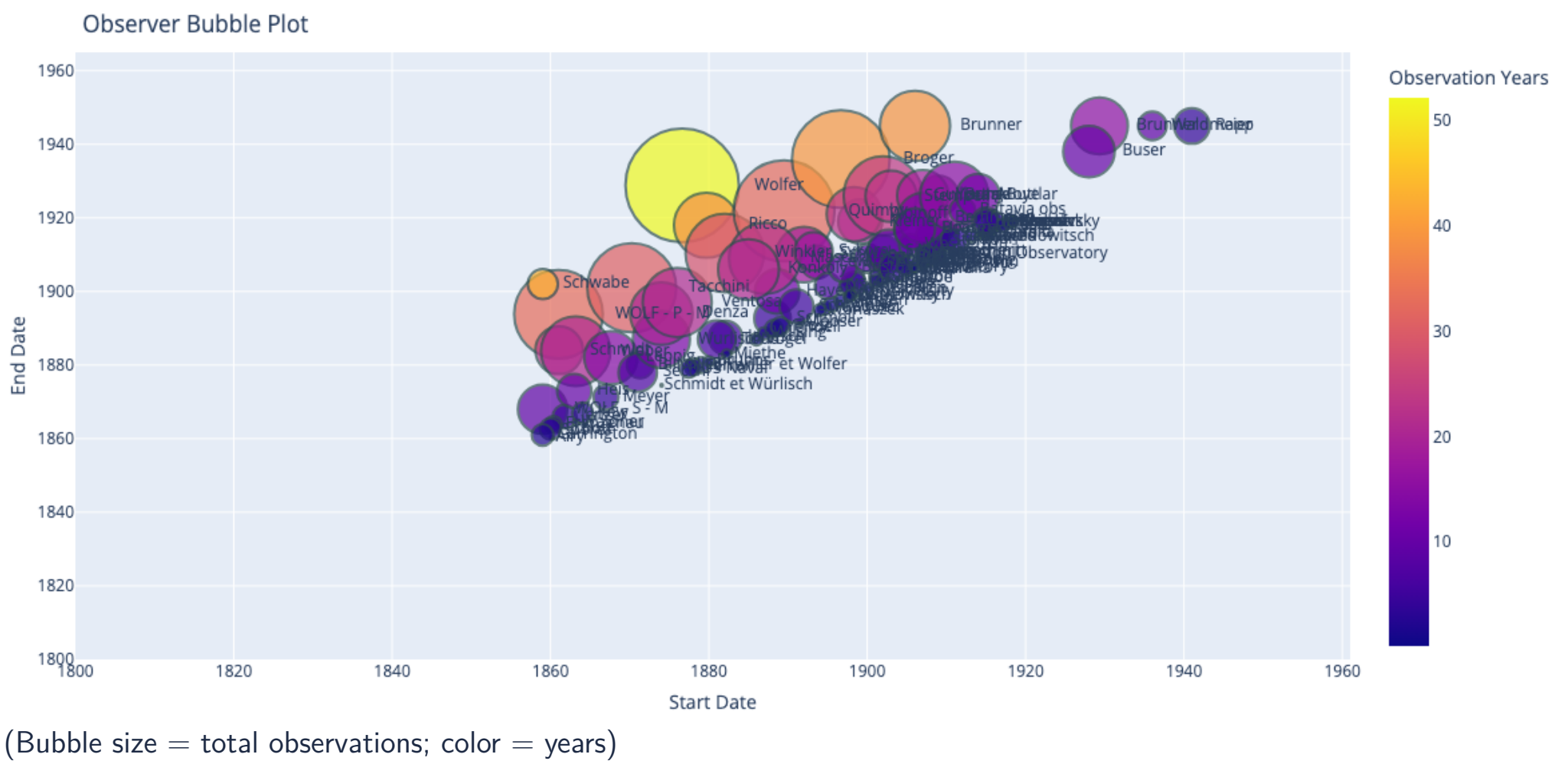
Source	Period	Current Focus
Zürich Tables	1945–1979	95% of 2000+ tables digitized; metadata encoding for late-1970s observers underway.
Mittheilungen	1610–1918	Fully integrated; observer statistics produced from 400+ years of entries.
Gruithuisen Manuscripts	1817–1849	25% transcription via Transkribus; file renaming algorithm + dual QC for group/spot counts.
C. H. Adams Drawings	1819–1823	338 sketches extracted (1056 entries); spotless days cross-checked against Wolf series.
Augustin Stark Ledgers	1813–1835	Normalizing tabular layouts; linking weather annotations to group counts.
Rost Journals	18th–19th c.	Building metadata vocabularies prior to digitization ingest.



Progress Highlights (Early 2025)

- Citizen-science ready interface blueprint finished; aligns with planned public transcription portal.
- Zürich production line blends Transkribus-assisted extraction with manual review to hit 2025 completion.
- Gruithuisen workflow pairs Kurrent script specialists with validation scripts to flag missing sun symbols.
- Adams drawings compared against *Mittheilungen* daily records to benchmark spotless-day agreement.

Observer Timeline



Methods

- Dual-mode transcription: tailored Transkribus handwriting models plus manual keying for complex tables.
- Data hygiene: observer alias reconciliation, calendar conversions, and spot/group recount audits.
- Uncertainty modelling: propagate observer calibration curves and note spotless-day confidence flags.
- Automation: reproducible Python pipelines with checkpoints for citizen-science ingestion and review.

Quality Gates

- Two-person verification for Adams and Gruithuisen counts before publishing derived indices.
- Cross-compare FARSuN transcriptions with Wolf's *Mittheilungen* and modern SILSO indices for drift detection.
- Metadata harmonisation ensures unique observer IDs, instrument notes, and weather context survive downstream.
- Git-tracked pipelines and notebooks capture every transformation for audit and reuse.

From Raw to ISN V3

- Curate daily counts with provenance-rich metadata (observer, instrument, weather context).
- Solve observer response functions to align group/spot scales across centuries.
- Aggregate to monthly, yearly, and hemispheric indices with propagated uncertainties.
- Benchmark against ISN V2 and satellite-era proxies to validate drift corrections.

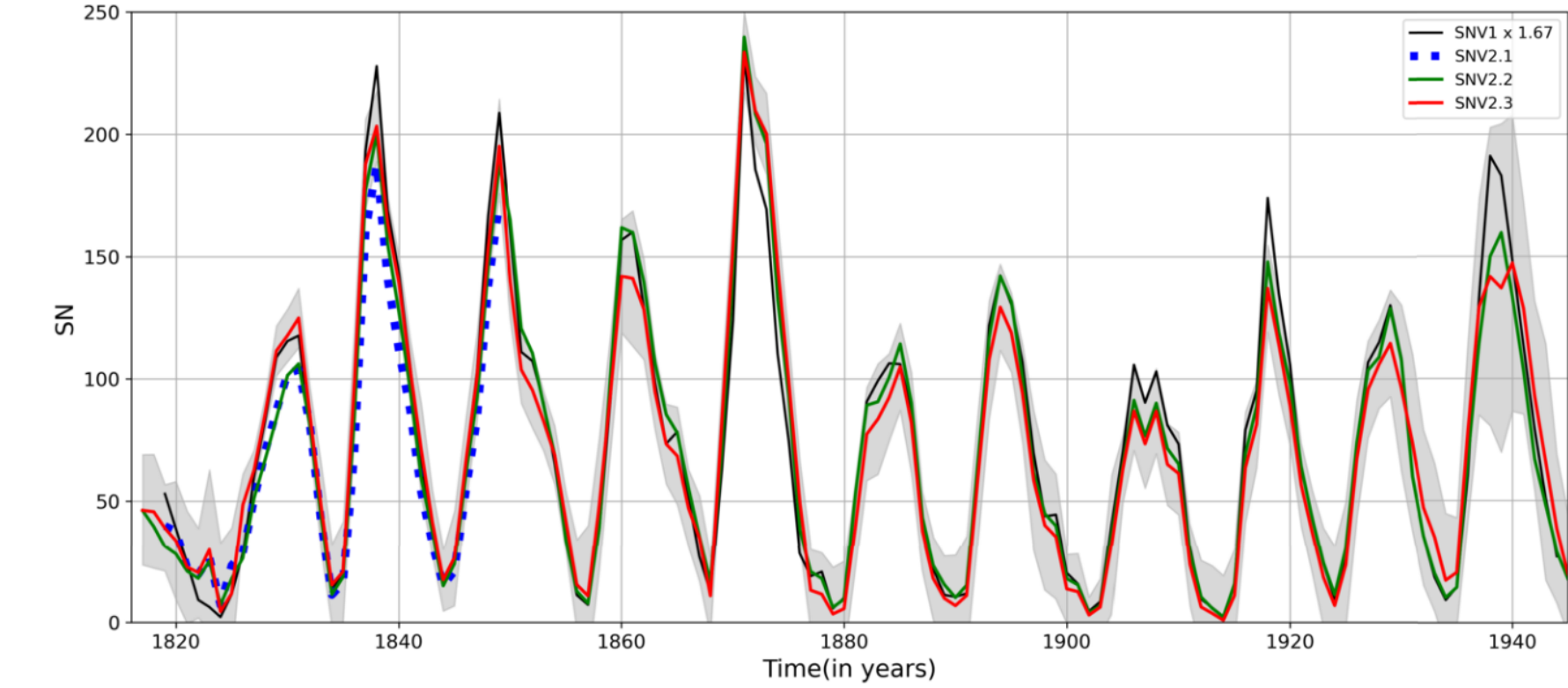


Figure 15 Annually averaged SN series. The grey-shaded area represents the 1 σ uncertainty of SNV2.3. We note that SNV1 has been multiplied by 1.67 to bring it to the same scale as the other series.

Challenges

- Deciphering *Kurrent* manuscripts and annotating non-solar notes without losing context.
- Harmonising heterogeneous formats (drawings, tabular ledgers, weather marginalia).
- Quantifying observer bias and spotless-day reporting habits over centuries.
- Maintaining end-to-end provenance from scanned folio to released data product.

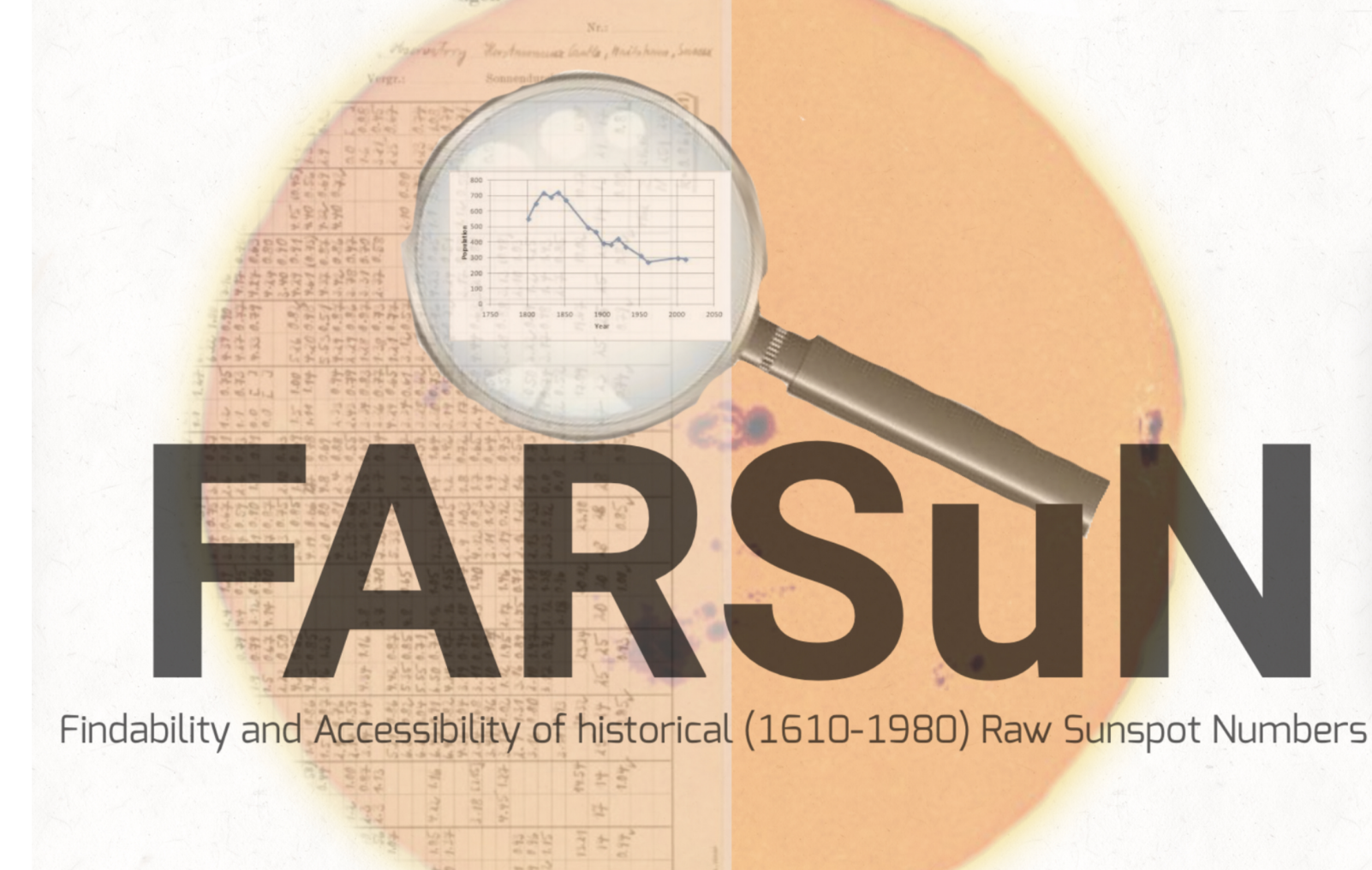
Space Climate Impact

Citizen science	Forecasting	Climate links	Open science
Interactive portal mobilises volunteers to transcribe <i>Kurrent</i> manuscripts and sustain STEM outreach.	Bias-free ISN V3 improves solar cycle prediction skill and operational space weather baselines.	Validated historical forcing refines long-term climate models and Earth radiation budget studies.	FAIR-compliant releases with APIs and uncertainty bundles accelerate community reuse.

Outlook

- Launch citizen-science transcription portal with tailored training sets.
- Publish ISN V3 release notes with uncertainty budget and API endpoints.
- Expand collaborations (Nagoya, UCLouvain, ROB) for cross-validation and joint publications.

Findability and Accessibility of historical (1610-1980) Raw Sunspot Numbers



References